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1. INTRODUCTION

1.1 Introduction:

Ethylene (IUPAC name: **ethene**) is a hydrocarbon (C_2H_4 or $H_2C=CH_2$).

Ethylene is widely used in the chemical industry and its worldwide production (over 225 millions tonnes in 2022) exceeds that of any other organic compound.

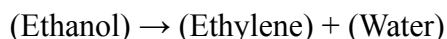
Ethylene is one of the most important petrochemical feedstocks in the world. It serves as a fundamental building block for the production of polymers, solvents, and various industrial chemicals. Traditionally, ethylene is produced from petroleum-based feedstocks such as naphtha and ethane through steam cracking. However, due to increasing environmental concerns, depletion of fossil fuels, and the need for sustainable development, renewable routes for ethylene production have gained significant importance.

One such sustainable method is the **dehydration of bioethanol**. Bioethanol is produced from renewable biomass sources such as sugarcane, corn, and agricultural residues. Converting bioethanol into ethylene provides a green and environmentally friendly alternative to conventional petrochemical routes.

The dehydration process involves the catalytic removal of water from ethanol to produce ethylene. The process is simple, requires fewer separation steps compared to steam cracking, and operates at comparatively lower temperatures.

1.1.1 Basic Reaction:

The primary reaction involved in ethanol degradation is:



The reaction is **catalytic** in nature, **Endothermic** and conducted in **vapor phase** inside **fixed or fluidized bed reactors** with catalysts at moderate pressures. The process in the fixed-bed reactor can be either isothermal or adiabatic, while it is usually adiabatic in the fluidized bed reactor.

The highest selectivity towards ethylene is obtained at **300–500 °C**. Higher temperatures shift the reaction towards acetaldehyde production, while lower temperatures result in production of diethyl ether.

1.2 Historical Development:

The dehydration of ethanol to produce ethylene was first practiced in the early 20th century before the dominance of petroleum-based steam cracking. During World War periods, ethanol dehydration gained importance due to shortages of petrochemical feedstocks.

With the advancement of petrochemical industries, steam cracking became dominant because of lower production cost at large scale. However, in recent decades, the bio-based route has regained importance due to rising crude oil prices, climate change concerns, Government policies promoting renewable chemicals, development of bio-refineries.

1.3 Applications of Ethylene:

Table 1.1: Uses of ethylene

APPLICATION	PRODUCT	GRADE	PURITY
Plastics	Polyethylene (LDPE, HDPE)	Polymer Grade	>99.9%
Antifreeze	Ethylene Glycol	Chemical Grade	99.5-99.9%
Packaging	Films and Containers	Polymer Grade	>99.9%
PVC Industry	Vinyl Chloride (PVC)	Polymer Grade	>99.9%
Chemical Intermediates	Ethylene Oxide (EO)	Chemical Grade	99.0-99.5%
Detergents	Linear Alpha Olefins	Polymer Grade	>99.9%
Agriculture	Promotes ripening of Fruits, Plants	Technical Grade	95-98%
Polymers	Ethylene-propylene rubber (EPR) for automotive parts	Polymer Grade	>99.9%
Pharmaceuticals	Monoethylene-glycol (MEG) and diethylene-glycol (DEG)	Polymer Grade	>99.9%

Bio-based ethylene enables production of **green polyethylene**, which is biodegradable and environmentally sustainable.

1.4 Properties of Ethylene:

1.4.1 Physical Properties:

Ethylene is colourless, flammable gas with a faint “sweet and musky” odour when pure. It is highly reactive due to its double bond.

Table 1.2: Physical Properties

Property	Symbol	Ethylene	Ethanol	Water
Molecular Formula	-	C ₂ H ₄	C ₂ H ₆ O	H ₂ O
Molecular Weight	MW	28.05 g/mol	46.07 g/mol	18 g/mol
Boiling Point	T _b	-103.7°C	78.37°C	100°C
Melting Point	T _m	-169.2°C	-114.1°C	0°C
Critical Temperature	T _c	9.2°C	241.1°C	374.1°C
Critical Pressure	P _c	50.4 atm	61.4 atm	218 atm
Critical Volume	V _c	131 cm ³ /mol	168 cm ³ /mol	56 cm ³ /mol
Density (at 25°C)	ρ	1.18 kg/m ³	789 kg/m ³	997 kg/m ³
Dynamic Viscosity	μ	9.5 × 10 ⁻⁶ Pa.s	1.07 × 10 ⁻³ Pa.s	0.89 × 10 ⁻³ Pa.s
Thermal Conductivity	k	0.016 W/m.K	0.171 W/m.K	0.598 W/m.K
Specific Heat Capacity	C _p	1.43 KJ/Kg.K	2.44 KJ/Kg.K	4.18 KJ/Kg.K
Heat of Vaporization	ΔH _{vap}	13.5 KJ/mol	38.6 KJ/mol	40.7 KJ/mol
Std. Enthalpy of Formation	ΔH _f ^o	+52.3 KJ/mol	-277 KJ/mol	-285.8 KJ/mol
Std. Gibbs Free Energy	ΔG _f ^o	+68.1 KJ/mol	-174.8 KJ/mol	-237.1 KJ/mol

1.4.2 Chemical Properties:

Unsaturated Hydrocarbon: The double bond is the site of reaction where other molecules add across it, breaking the π bond to become saturated.

Property of Ethylene	Description
Hydrogenation	Reacts with Hydrogen in presence of catalyst to form ethane
Halogenation	Reacts with halogens to form dihaloethanes
Hydration	Reacts with water in presence of acid catalyst to form ethanol
Hydrohalogenation	Reacts with halides to form ethylhalides
Polymerization	Under high temperature and pressure, ethene molecules polymerize to form polyethene
Combustion	$C_2H_4 + 3O_2 \rightarrow 2CO_2 + 2H_2O$
Oxidation	Readily oxidized by oxidizing agents to form ethylene glycol
Ozonolysis	Reacts with ozone to form an ozonide, which upon reduction yields formaldehyde
Solubility	Insoluble in Water, soluble in organic solvents

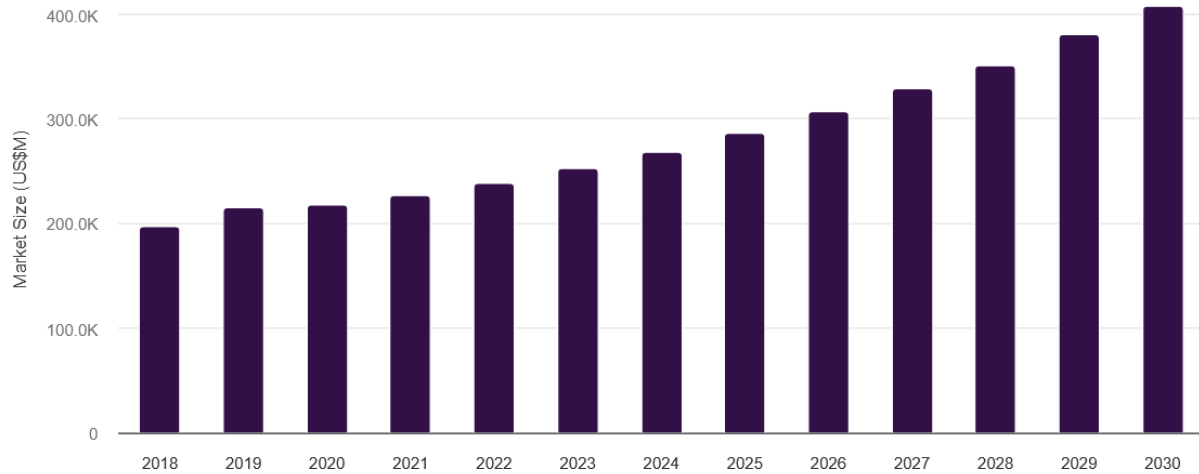
1.5 Demand and Supply:

Global ethylene demand, driven heavily by polyethylene for packaging and construction, reached approximately **184–185 million metric tons (MMT)** in 2024–2025. The market is expected to grow at a **CAGR of 5.8% to 6.9%** through 2030, with demand potentially exceeding 400 MMT by 2030–2032. India produced about 5 million Metric tonnes in 2024-2025.

Below is some data about Indian Ethylene market demand-

Market Size Volume in FY2023	7.4 Million Tonnes
Market Size Volume in FY2030	12.5 Million Tonnes

Growth Rate	CAGR of 6.12% from FY2024 to FY2030
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Global ethylene market, 2018-2030 (US\$M):**Fig 1.1 Global Ethylene Market****1.6 Aim and Objective:**

The objectives of this project are:

1. To study the dehydration of bioethanol for ethylene production.
2. To develop a process flow diagram (PFD).
3. To analyze the thermodynamic feasibility of the reaction.
4. To perform material balance calculations.
5. To calculate energy requirements of the process.
6. To design major process equipment such as reactors and separation units.
7. To evaluate environmental and safety aspects.
8. To estimate plant cost and layout considerations.

2. LITERATURE SURVEY

2.1 Method of Manufacture:

Bioethanol dehydration is the most widely adopted catalytic route for the production of polymer-grade ethylene from renewable feedstocks. The process involves the acid-catalysed intramolecular dehydration of ethanol over a solid catalyst, typically alumina (Al_2O_3) or zeolites such as HZSM-5, at temperatures ranging from 300°C to 500°C. This route has gained significant industrial traction due to the abundance of bioethanol from fermentation of lignocellulosic and starchy biomass, and the increasing global demand for bio-based ethylene as a sustainable alternative to naphtha cracking.

The stoichiometry of the primary reaction is: $\text{C}_2\text{H}_5\text{OH} \rightarrow \text{C}_2\text{H}_4 + \text{H}_2\text{O}$ ($\Delta H = +45.5$ kJ/mol). The reaction is endothermic and proceeds most favourably at elevated temperatures above 350°C at moderate pressures. Side reactions include the formation of diethyl ether (DEE) at lower temperatures (below 300°C) and acetaldehyde or coke at excessively high temperatures. Selectivity to ethylene is therefore strongly dependent on temperature, space velocity, and catalyst choice.

The selected process for this minor project employs an adiabatic fixed bed reactor operating at 425°C and 4 bar with a $\gamma\text{-Al}_2\text{O}_3$ catalyst, achieving a single-pass conversion of 87% at a selectivity exceeding 99% towards ethylene. Feed preheating is accomplished through a two-stage feed-effluent heat exchanger (FEHE) network, followed by an electric resistance heater for final superheating, which significantly reduces the external energy demand of the process. The process flow sheet for the designed plant is shown in Figure 2.1.

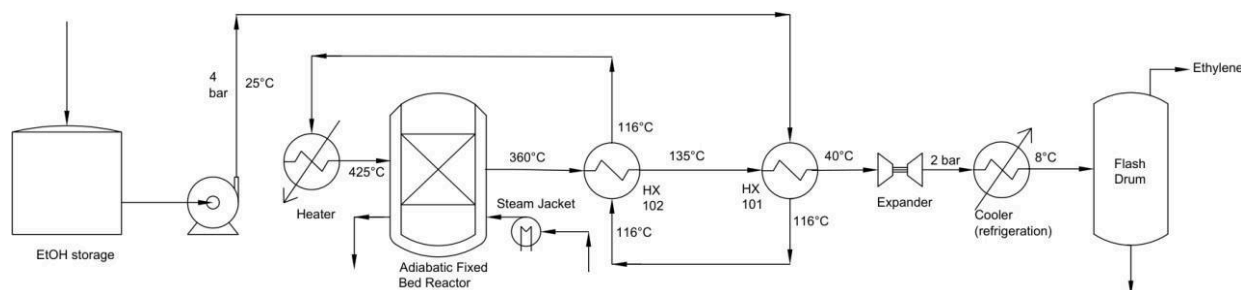


Figure 2.1: Process flow sheet for bioethanol dehydration to ethylene.

2.2 Process Description:

The overall process can be broadly categorised into four sections: feed preparation,

reaction, heat recovery, and product separation. Each section is discussed below along with the design decisions and modifications relevant to the present plant.

2.2.1 Feed Preparation and Preheating:

The ethanol feed is stored as liquid at ambient conditions and is pressurised to 4 bar using a centrifugal pump before entering the heat integration network. The preheating of the feed is accomplished in two stages. In the first stage, the feed enters Heat Exchanger HX 101, where it is preheated from 25°C to approximately 116°C by exchanging heat with the partially cooled reactor product stream (135°C → 40°C). In the second stage, the feed is partially vaporized (45% vaporization) in HX 102 by the hot reactor effluent entering at 360°C and leaving at 135°C. This counter-current cascade configuration maximises heat recovery and substantially reduces the duty of the downstream electric heater.

Following the two-stage preheat, the feed enters an electric resistance heater that fully vaporizes the feed and superheats it from 116°C to the reactor inlet temperature of 425°C. Electric heating was selected over a fired furnace to allow precise temperature control and eliminate combustion-related emissions at the lab/pilot scale of this project. The use of electric heating also aligns with the green chemistry principles underlying bioethanol-based ethylene production.

2.2.2 Reactor:

The reactor selected is an adiabatic fixed bed reactor packed with γ -Al₂O₃ catalyst, operated at 425°C and 4 bar. Alumina catalysts are well-established for ethanol dehydration and offer high selectivity to ethylene at temperatures above 350°C, with minimal coking compared to zeolite-based catalysts at equivalent conditions [1]. The adiabatic configuration was chosen for its simplicity and low capital cost, though the endothermic nature of the reaction leads to a temperature drop across the bed, which is accounted for in the design by operating the inlet temperature sufficiently above the minimum required reaction temperature.

A steam jacket is incorporated around the reactor shell. This serves dual purposes: it prevents excessive heat loss from the reactor body to the surroundings, maintaining near-adiabatic conditions, and the generated steam can be utilised for process utilities. A circulation pump ensures continuous steam flow through the jacket. The reactor achieves a conversion of 87% on a single pass basis, with ethylene selectivity exceeding 99%, as side reactions such as DEE formation are suppressed at the operating temperature of 425°C.

2.2.3 Heat Recovery Section:

The hot reactor effluent exits at 360°C and is cooled through the two-stage FEHE (Feed Effluent Heat Exchange) network described in Section 2.2.1, in the order HX 102 (360°C → 135°C) followed by HX 101 (135°C → 40°C). This counter-current arrangement ensures that the highest-grade heat from the reactor outlet is utilised to heat the feed to its highest preheat temperature in HX 102, while the lower-grade heat in HX 101 provides initial preheating of the cold feed.

Following heat recovery, the cooled stream at 40°C is passed through an expander that reduces the pressure from 4 bar to 2 bar. The expander was selected over a simple throttle valve or ejector to recover the pressure energy as useful mechanical work, improving the overall energy efficiency of the process. The pressure reduction across the expander also causes an associated temperature drop, which assists the subsequent refrigeration duty.

2.2.4 Separation Section:

Following the expander, the stream is cooled to 8°C in a refrigeration cooler. The low temperature facilitates condensation of water and any unreacted ethanol, while ethylene remains in the vapour phase due to its very low boiling point (−104°C at 1 atm). The cooled two-phase mixture is fed to a flash drum operating at 2 bar and 8°C, where vapour-liquid equilibrium governs the separation. The vapour overhead product is polymer-grade ethylene at approximately 99.55% purity, while the liquid bottoms contain primarily water and trace unreacted ethanol.

The flash separation was analysed using the Rachford-Rice equation to determine the vapour fraction and component distribution between phases.

2.3 Catalyst Selection:

The choice of catalyst is critical to the performance of the dehydration reactor. Two main categories of catalysts are reported in literature: alumina-based (Al_2O_3) and zeolite-based (HZSM-5, H-Beta). Alumina catalysts are preferred for ethylene production at temperatures of 350–500°C due to their moderate acidity, which promotes dehydration while minimising cracking and coking. HZSM-5 zeolites offer higher activity at lower temperatures but are prone to coke deposition and require periodic regeneration [2]. For the present design, $\gamma\text{-Al}_2\text{O}_3$ was selected for its well-established industrial track record, lower cost, and longer catalyst lifetime under the operating conditions.

2.4 Process Variants and Comparison:

The primary industrial route for ethylene production remains steam cracking of naphtha or ethane, which accounts for over 95% of global ethylene supply. However, bio-based routes via ethanol dehydration have grown in commercial significance, particularly in Brazil (Braskem's green polyethylene programme) and India, driven by ethanol availability from sugarcane and grain fermentation. A comparative assessment of the two routes reveals that while steam cracking offers much higher throughput and lower per-tonne production cost at large scale, ethanol dehydration offers a lower carbon footprint, simpler downstream separation (no BTX, no methane separation required), and compatibility with existing biorefinery infrastructure [3].

Within the ethanol dehydration family, two reactor configurations are widely reported: adiabatic multi-stage reactors with inter-stage reheating, and isothermal reactors with jacket heating. Multi-stage adiabatic configurations are preferred at industrial scale due to their scalability and ease of catalyst replacement. The single-stage adiabatic reactor with steam jacketing adopted in the present design is appropriate for the 100 TPA scale of this project.

2.5 Product Specification:

Polymer-grade ethylene requires a purity of ≥ 99 mol% for use in polymerisation processes. Pipeline or chemical-grade ethylene typically requires ≥ 98 mol% purity. The present process targets a flash drum overhead product of 99.38 mol% ethylene purity, placing it in the chemical-grade category and suitable for downstream polyethylene synthesis after minor further purification if required. The primary impurity in the product stream is water vapour, which can be removed by molecular sieve drying downstream of the flash drum.

Using the bioethanol dehydration process, products of high purity can be obtained provided the ethanol feedstock itself is of sufficient grade. The theoretical maximum Ethylene production based on stoichiometry is 15.21 kg from 24.985 kg of ethanol. The actual ethylene obtained is 12.519 kg, giving an overall Ethylene yield of 82.3%.

3. THERMODYNAMIC FEASIBILITY

3.1 Purpose of Thermodynamic Calculations:

Thermodynamic calculations are very important as they allow us to determine, in the preliminary stage, whether further evaluation of the project should be carried out or not. The feasibility and conditions for the process are derived on the basis of energy, enthalpy, entropy and Gibbs free energy. For the dehydration of bioethanol to ethylene, the thermodynamic feasibility must be demonstrated at the chosen reactor operating temperature of 425°C (698 K) and 4 bar before any design or simulation work can be justified.

3.2 Heat of Reaction:

If a process involves a chemical reaction, heat will normally have to be added or removed. The amount of heat given out or absorbed in a chemical reaction depends on the conditions under which the reaction is carried out. The standard heat of any reaction can be calculated from the heats of formation of the products and reactants using the following relationship:

$$\Delta H^{\circ}_r = \sum \Delta H^{\circ}_f, \text{Products} - \sum \Delta H^{\circ}_f, \text{Reactants} \quad \text{---(3.1)}$$

The dehydration of ethanol to ethylene proceeds according to the following reaction:



The standard enthalpies of formation of all reactants and products involved in the dehydration reaction at 25°C are tabulated in Table 3.1:

Table 3.1: Standard Enthalpy of Formation at 25°C (298 K)

Component	Formula	$\Delta H^{\circ}_f, 298$ (kJ/mol)	Phase
Ethanol	C ₂ H ₅ OH	−235.3	Vapour

Component	Formula	$\Delta H^\circ_f, 298$ (kJ/mol)	Phase
Ethylene	C_2H_4	+52.5	Vapour
Water	H_2O	-241.8	Vapour

3.2.1 Standard Heat of Reaction at 25°C (298 K):

Applying equation (3.1) to the dehydration reaction:

$$\Delta H^\circ_{r,298} = [\Delta H^\circ_f(C_2H_4) + \Delta H^\circ_f(H_2O)] - [\Delta H^\circ_f(C_2H_5OH)]$$

$$\Delta H^\circ_{r,298} = [52.5 + (-241.8)] - [-235.3] \text{ kJ/mol}$$

$$\Delta H^\circ_{r,298} = (-189.3) - (-235.3) \text{ kJ/mol}$$

$$\Delta H^\circ_{r,298} = +46.0 \text{ kJ/mol} \quad \text{---(3.4)}$$

ΔH_r is positive; hence the reaction is endothermic at standard conditions.

3.2.2 Heat of Reaction at 425°C (698 K) using Kirchhoff's Law:

Since the reactor operates at 425°C, the standard heat of reaction must be corrected to the operating temperature using Kirchhoff's Law:

$$\Delta H^\circ_{r,T} = \Delta H^\circ_{r,298} + \int \Delta C_p dT \quad \text{---(3.4)}$$

Where ΔC_p is the difference in molar heat capacities of products and reactants. The mean molar C_p values used are tabulated in Table 3.2:

Table 3.2: Mean Molar Heat Capacities (298 K $\rightarrow$ 698 K)

Component	C_p (J/mol·K)	Role
Ethanol (g)	65.5	Reactant
Ethylene (g)	42.9	Product
Water (g)	33.6	Product

- $\Delta C_p = C_p(C_2H_4) + C_p(H_2O) - C_p(C_2H_5OH)$

- $\Delta C_p = 42.9 + 33.6 - 65.5 = +11.0 \text{ J/mol}\cdot\text{K}$
- Temperature correction: $\Delta T = 698 - 298 = 400 \text{ K}$
- $\int \Delta C_p dT = 11.0 \times 400 = 4,400 \text{ J/mol} = +4.4 \text{ kJ/mol}$

$$\Delta H_{r,698} = 46.0 + 4.4 = +50.4 \text{ kJ/mol} \quad \text{---(3.5)}$$

ΔH_r is positive at 425°C; the reaction remains endothermic at reactor operating conditions. An external heat source (electric heater) is therefore required to supply the heat of reaction.

3.3 Thermodynamic Feasibility:

The Gibbs Free Energy predicts the feasibility and equilibrium condition of a reaction at constant temperature and pressure. If the change in Gibbs Free Energy (ΔG) is negative, the thermodynamics of the reaction is favourable. If ΔG is highly positive, the reaction is thermodynamically not feasible. This can be evaluated using the relation:

$$\Delta G = \Delta H - T\Delta S \quad \text{---(3.6)}$$

The standard entropy values for all species at 25°C are given in Table 3.3, from which the standard entropy change of reaction is calculated:

Table 3.3: Standard Entropy of Formation at 25°C (298 K)

Component	S°298 (J/mol·K)	Phase
Ethanol (g)	282.7	Vapour
Ethylene (g)	219.3	Vapour
Water (g)	188.7	Vapour

3.3.1 Standard Entropy Change of Reaction:

$$\Delta S^{\circ}_{\text{rxn}} = [S^{\circ}(\text{C}_2\text{H}_4) + S^{\circ}(\text{H}_2\text{O})] - [S^{\circ}(\text{C}_2\text{H}_5\text{OH})]$$

$$\Delta S^{\circ}_{\text{rxn}} = [219.3 + 188.7] - [282.7]$$

$$\Delta S^{\circ}_{\text{rxn}} = +125.3 \text{ J/mol}\cdot\text{K} \quad \text{---(3.7)}$$

ΔS is positive; the reaction produces more moles of gas than it consumes (1 mol reactant $\rightarrow$ 2 mol products), increasing molecular disorder. This is thermodynamically favourable.

3.3.2 Gibbs Free Energy at 425°C (698 K):

Using the corrected enthalpy at 698 K and the standard entropy change, the Gibbs Free Energy at reactor operating conditions is calculated:

$$\Delta G_{698} = \Delta H^{\circ}_{r,698} - T\Delta S^{\circ}_{rxn}$$

$$\Delta G_{698} = 50,400 - (698 \times 125.3) \text{ J/mol}$$

$$\Delta G_{698} = 50,400 - 87,459 \text{ J/mol}$$

$$\Delta G_{698} = -37,059 \text{ J/mol} = -37.06 \text{ kJ/mol} \quad \text{---(3.8)}$$

ΔG is negative at 425°C; hence the dehydration reaction is thermodynamically feasible at the chosen reactor operating conditions.

3.4 Equilibrium Constant:

The equilibrium constant K provides a quantitative measure of the extent to which the reaction proceeds to completion at the operating temperature. It is related to the Gibbs Free Energy by:

$$\Delta G = -RT \ln K \quad \text{---(3.9)}$$

$$\ln K = -\Delta G / RT = 37,059 / (8.314 \times 698)$$

$$\ln K = 37,059 / 5,803.2 = 6.386$$

$$K = e^{6.386} = 593.477 \quad \text{---(3.10)}$$

$K \gg 1$ at 425°C; the equilibrium strongly favours the products, confirming that high ethylene conversion is achievable at the reactor operating temperature.

3.5 Effect of Temperature on Feasibility:

Since the dehydration reaction is endothermic ($\Delta H > 0$) and produces an increase in entropy ($\Delta S > 0$), the sign of ΔG is determined by the relative magnitudes of the enthalpy and entropy terms. At low temperatures, the positive enthalpy term dominates and ΔG is positive, making the reaction thermodynamically infeasible. As temperature increases, the $T\Delta S$ term grows and eventually exceeds ΔH , causing ΔG to become negative.

The crossover temperature at which $\Delta G = 0$ can be estimated as:

$$T^* = \Delta H / \Delta S = 46,000 / 125.3 \approx 367 \text{ K} \approx 94^\circ\text{C} \quad \text{---(3.11)}$$

Above approximately 94°C , the reaction becomes thermodynamically spontaneous. The reactor operating temperature of 425°C is well above this threshold, ensuring a strongly negative ΔG and high equilibrium conversion. This also explains why the reaction requires elevated temperatures in practice — at lower temperatures both thermodynamic feasibility and reaction kinetics are unfavourable, and the catalyst selectivity towards diethyl ether (DEE) formation increases.

3.6 Summary of Thermodynamic Calculations:

The key thermodynamic parameters for the ethanol dehydration reaction at the reactor operating conditions of 425°C and 4 bar are summarised in Table 3.4:

Table 3.4: Summary of Thermodynamic Parameters at 425°C (698 K)

Parameter	Symbol	Value	Unit	Inference
Standard heat of rxn (25°C)	$\Delta H^\circ_{r,298}$	+46.0	kJ/mol	Endothermic
Heat of rxn at 425°C	$\Delta H^\circ_{r,698}$	+50.4	kJ/mol	Endothermic
Entropy change	$\Delta S^\circ_{\text{rxn}}$	+125.3	J/mol·K	Favourable
Gibbs Free Energy at 425°C	ΔG_{698}	-37.06	kJ/mol	Feasible

Parameter	Symbol	Value	Unit	Inference
Equilibrium constant at 425°C	K	596	—	Product-favoured
Crossover temperature (T*)	T*	~94	°C	Min. feasibility temp.

The thermodynamic analysis confirms that the dehydration of bioethanol to ethylene is thermodynamically feasible at 425°C, with a strongly negative Gibbs Free Energy of –37.06 kJ/mol and an equilibrium constant of 596. The positive entropy change arising from the increase in the number of moles of gas upon reaction provides a strong thermodynamic driving force at elevated temperatures. The chosen operating temperature of 425°C is well justified on thermodynamic grounds and is consistent with reported industrial and laboratory-scale conditions for alumina-catalysed ethanol dehydration.

4. MATERIAL BALANCE

4.1 Purpose for Material Balance:

Material balances are the basis of process design. A material balance taken over the complete process will determine the quantities of raw materials required and products produced. Balances over individual process units set the process stream flows and compositions. Material balances are also useful tools for the study of plant operation and trouble shooting. The objective of this section is to determine all the process streams and their compositions for the bioethanol-to-ethylene dehydration plant with a capacity of 100 TPA ethylene. The balance is performed over the entire process and over individual equipment using the DWSIM simulation data as the basis.

4.2 Overall Material Balance:

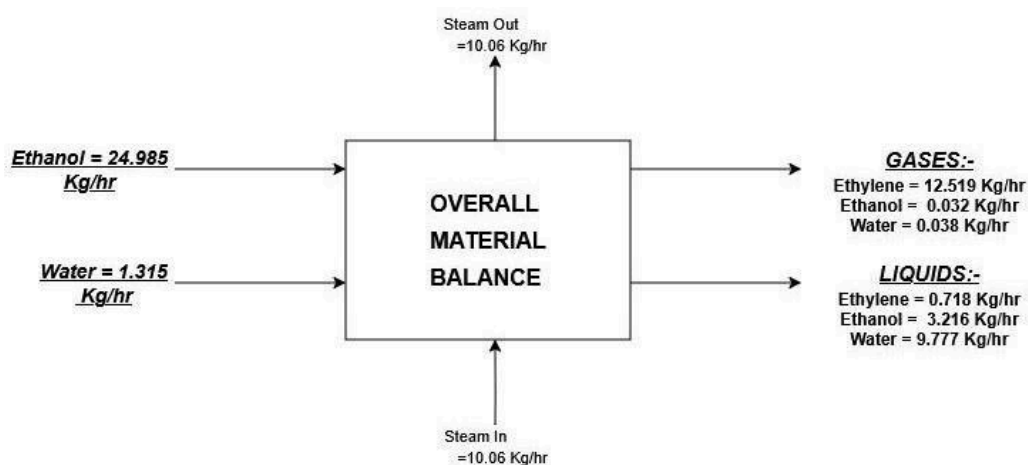
Basis: 26.3 kg/hr total feed (95 wt% ethanol, 5 wt% water) at 1 bar, 25°C, with 87% single-pass conversion in the adiabatic fixed bed reactor.

- Total feed flow rate = 26.3 kg/hr
- Ethanol in feed = $26.3 \times 0.95 = 24.985$ kg/hr
- Water in feed = $26.3 \times 0.05 = 1.315$ kg/hr
- Ethanol reacted = $24.985 \times 0.87 = 21.737$ kg/hr
- Ethanol unreacted = $24.985 \times 0.13 = 3.248$ kg/hr
- Moles of ethanol reacted = $21.737 \times 1000 / 46 = 472.5$ mol/hr
- Ethylene produced (1:1 stoichiometry) = $472.5 \times 28 / 1000 = 13.23$ kg/hr
- Water produced by reaction = $472.5 \times 18 / 1000 = 8.505$ kg/hr
- Total water in product stream = $1.315 + 8.505 = 9.820$ kg/hr

Table 4.1: Overall Material Balance across the Plant

Component	Inlet (kg/hr)	Outlet (kg/hr)
Ethanol	24.985	3.248

Component	Inlet (kg/hr)	Outlet (kg/hr)
Water	1.315	9.815
Ethylene	0	13.237
Total	26.300	26.300



4.3 Material Balance Across Individual Equipment:

The material balance across individual components of the flowsheet is presented below.

4.3.1 Feed Pump:

Table 4.2: Material Balance across Feed Pump

Component	Inlet (kg/hr)	Outlet (kg/hr)
Ethanol	24.985	24.985
Water	1.315	1.315
Ethylene	0	0
Total	26.300	26.300

Conditions: Inlet — 1 bar, 25°C | Outlet — 4 bar, 25°C | No composition change across pump.

4.3.2 Heat Exchanger HX 101 (Feed Preheat — Cold Side):

Table 4.3: Material Balance across HX 101

Component	Cold Inlet (kg/hr)	Cold Outlet (kg/hr)	Hot Inlet (kg/hr)	Hot Outlet (kg/hr)
Ethanol	24.985	24.985	3.248	3.248
Water	1.315	1.315	9.815	9.815
Ethylene	0	0	13.237	13.237
Total	26.300	26.300	26.300	26.300

Cold side: 25°C → 116°C, 4 bar, liquid (VF = 0). Hot side: 135°C → 40°C, 4 bar.

4.3.3 Heat Exchanger HX 102 (Partial Vaporisation — Cold Side):

Table 4.4: Material Balance across HX 102

Component	Cold Inlet (kg/hr)	Cold Outlet (kg/hr)	Hot Inlet (kg/hr)	Hot Outlet (kg/hr)
Ethanol	24.985	24.985	13.237	13.237
Water	1.315	1.315	9.815	9.815
Ethylene	0	0	3.248	3.248
Total	26.300	26.300	26.300	26.300

Cold side: 116°C → 116°C, 4 bar, VF = 0 → 0.45 (partial vaporisation). Hot side: 360°C → 135°C, 4 bar.

4.3.4 Electric Heater:

Table 4.5: Material Balance across Electric Heater

Component	Inlet (kg/hr)	Outlet (kg/hr)
Ethanol	24.985	24.985
Water	1.315	1.315
Ethylene	0	0
Total	26.300	26.300

Inlet: 116°C, 4 bar, VF = 0.45 | Outlet: 450°C, 4 bar, VF = 1 (fully superheated vapour).

4.3.5 Adiabatic Fixed Bed Reactor:

Table 4.6: Material Balance across Reactor

Component	Inlet (kg/hr)	Outlet (kg/hr)	Remarks
Ethanol	24.985	3.248	87% converted
Water	1.315	9.815	Reaction produces water
Ethylene	0	13.237	Product formed
Total	26.300	26.300	~Balance

Inlet: 450°C, 4 bar | Outlet: 360°C, 4 bar | Reaction: $\text{C}_2\text{H}_5\text{OH} \rightarrow \text{C}_2\text{H}_4 + \text{H}_2\text{O}$, conversion = 87%.

4.3.6 Expander:

Table 4.7: Material Balance across Expander

Component	Inlet (kg/hr)	Outlet (kg/hr)
Ethanol	3.248	3.248
Water	9.815	9.815
Ethylene	13.237	13.237
Total	26.300	26.300

Inlet: 40°C, 4 bar | Outlet: 28.43°C, 2 bar | No composition change.

4.3.7 Refrigeration Cooler:

Table 4.8: Material Balance across Refrigeration Cooler

Component	Inlet (kg/hr)	Outlet (kg/hr)
Ethanol	3.248	3.248
Water	9.815	9.815
Ethylene	13.237	13.237
Total	26.300	26.300

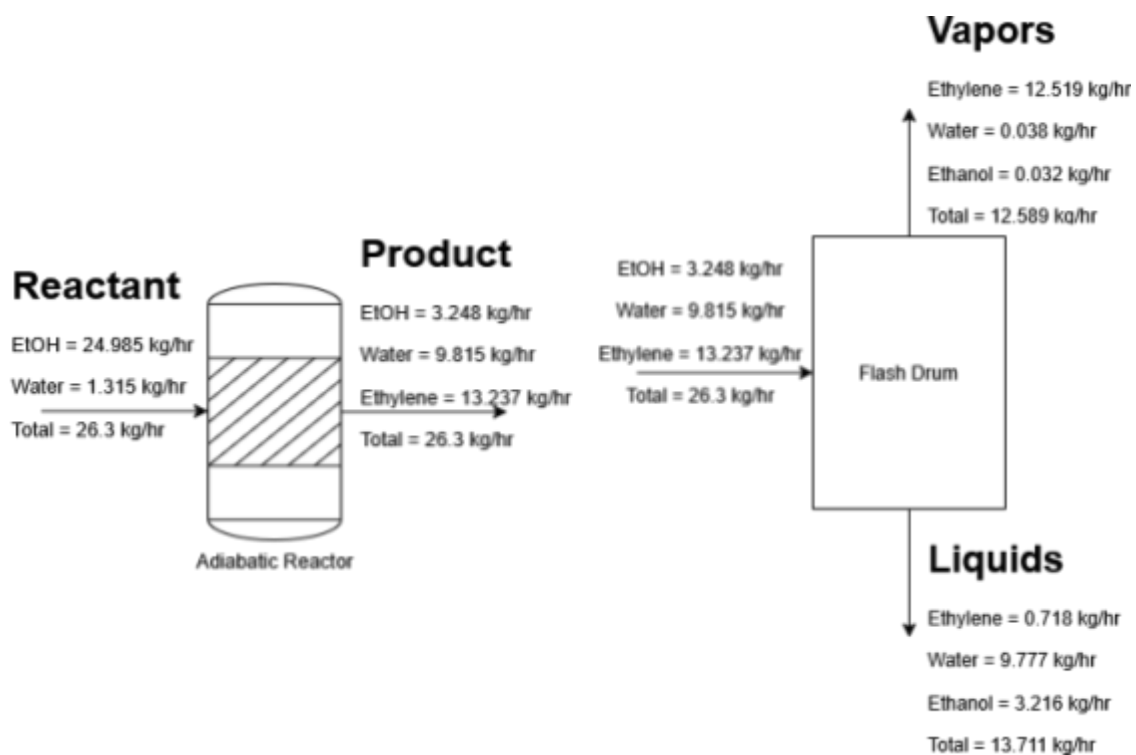
Inlet: 28.43°C, 2 bar | Outlet: 8.43°C, 2 bar | No composition change.

4.3.8 Flash Drum:

Table 4.9: Material Balance across Flash Drum

Component	Inlet (kg/hr)	Vapour Outlet (kg/hr)	Liquid Outlet (kg/hr)
Ethylene	13.237	12.519	0.718
Water	9.815	0.038	9.777
Ethanol	3.248	0.032	3.216
Total	26.300	12.589	13.711

Conditions: 8.43°C, 2 bar. Vapour (ethylene product): 99.44 wt% ethylene. Liquid (bottoms): primarily water and unreacted ethanol.



5. ENERGY BALANCE

5.1 Purpose for Energy Balance:

As with mass, energy can be considered to be separately conserved in all but nuclear processes. In process design, energy balances are made to determine the energy requirements of the process: the heating, cooling and power required. The steady state energy balance is generally calculated using the relation:

$$\text{Energy Out} = \text{Energy In} + \text{Generation} - \text{Consumption}$$

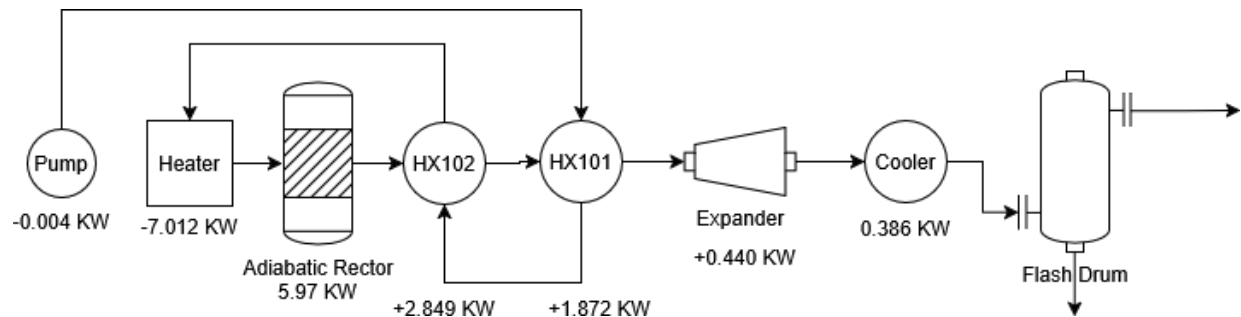
In this section, the enthalpy balance is performed over individual equipment to estimate the amount of utilities consumed by the plant for the production of 100 TPA ethylene via bioethanol dehydration. Heat losses to surroundings are assumed to be negligible and all pumps are assumed to neither supply nor dissipate significant heat. Literature values of specific heat capacities and latent heats are used throughout.

Basis:

- Total feed flow rate = 26.3 kg/hr = 7.306×10^{-3} kg/s
- Feed composition: 95 wt% ethanol, 5 wt% water
- Operating pressure: 4 bar (feed side), 2 bar (separation side)

Literature Cp values used:

- Cp ethanol (liquid) = 2.44 kJ/kg·K | Cp ethanol (vapour) = 1.42 kJ/kg·K
- Cp water (liquid) = 4.18 kJ/kg·K | Cp water (vapour) = 2.01 kJ/kg·K
- Cp ethylene (vapour) = 1.55 kJ/kg·K
- Latent heat ethanol (at 4 bar) λ = 800 kJ/kg
- Latent heat water (at 4 bar) λ = 2133 kJ/kg
- Heat of reaction (dehydration) ΔH_{rxn} = +45.5 kJ/mol (endothermic)



5.2 Enthalpy Balance across Feed Pump:

Formula: $W = \dot{m} \times \Delta P / (\rho \times \eta)$

- $\dot{m} = 26.3 / 3600 = 7.306 \times 10^{-3} \text{ kg/s}$
- $\Delta P = (4 - 1) \times 10^5 = 3 \times 10^5 \text{ Pa}$
- $\rho \text{ mixture (95\% ethanol-water, } 25^\circ\text{C)} = 789 \text{ kg/m}^3$
- Pump efficiency $\eta = 0.75$ (assumed)
- Volumetric flow $= 7.306 \times 10^{-3} / 789 = 9.26 \times 10^{-6} \text{ m}^3/\text{s}$

$$W_{\text{pump}} = (9.26 \times 10^{-6} \times 3 \times 10^5) / 0.75 = 3.70 \text{ W} \approx 0.004 \text{ kW}$$

Pump duty is negligibly small as expected for liquid pumping over a small pressure differential.

5.3 Enthalpy Balance across HX 101:

HX 101 preheats the liquid feed from 25°C to 116°C on the cold side, while cooling the reactor product from 135°C to 40°C on the hot side. At 4 bar, water condenses below $\sim 144^\circ\text{C}$ and ethanol condenses below $\sim 92^\circ\text{C}$. Since the hot side spans $135^\circ\text{C} \rightarrow 40^\circ\text{C}$, both water and ethanol are in the liquid phase throughout this range and liquid C_p values are applied. Additionally, 5.3% of ethanol undergoes partial condensation in the upper section of the exchanger, contributing latent heat.

Cold Side (Feed: $25^\circ\text{C} \rightarrow 116^\circ\text{C}$, liquid):

- $\Delta T_{\text{cold}} = 116 - 25 = 91^\circ\text{C}$

$$Q = \dot{m} \times C_p \times \Delta T$$

- $Q_{\text{ethanol}} = (24.985/3600) \times C_{p_liquid} \times \Delta T = (24.985/3600) \times 2.44 \times 91 = 1.733 \text{ kW}$
- $Q_{\text{water}} = (1.315/3600) \times C_{p_liquid} \times \Delta T = (1.315/3600) \times 4.18 \times 91 = 0.139 \text{ kW}$
 $Q_{\text{HX101 (cold)}} = 1.733 + 0.139 = 1.872 \text{ kW}$

Hot Side (Reactor product: 135°C → 40°C, liquid Cp applied):

- $\Delta T_{\text{hot}} = 135 - 40 = 95^\circ\text{C}$
- $Q = \dot{m} \times C_p \times \Delta T$
- $Q = \dot{m} \times \lambda$
- $Q_{\text{ethylene}} = (13.237/3600) \times 1.55 \times 95 = 0.542 \text{ kW}$ [vapour throughout]
- $Q_{\text{water}} = (9.815/3600) \times 4.18 \times 95 = 1.082 \text{ kW}$ [liquid Cp, condensed below 144°C]
- $Q_{\text{ethanol (sensible)}} = (3.248/3600) \times 2.44 \times 95 = 0.210 \text{ kW}$ [liquid Cp, condensed below 92°C]
- $Q_{\text{ethanol (latent, 5.3% partial condensation)}} = (0.053 \times 3.248/3600) \times 800 = 0.038 \text{ kW}$

$$Q_{\text{HX101 (hot)}} = 0.542 + 1.082 + 0.210 + 0.038 = 1.872 \text{ kW}$$

Heat balance check: Cold side = 1.872 kW, Hot side = 1.872 kW → Exact balance

5.4 Enthalpy Balance across HX 102:

HX 102 partially vaporises the feed at 116°C (bubble point at 4 bar) from VF = 0 to VF = 0.45. The cold side absorbs latent heat at approximately constant temperature. On the hot side, the reactor product cools from 360°C to 135°C. As the stream approaches 135°C it is near the water dew point at 4 bar (~144°C), causing 0.7% of the water to begin condensing, releasing additional latent heat which exactly balances the cold side requirement.

Cold Side (Feed: 116°C → 116°C, VF: 0 → 0.45, latent heat):

- Ethanol vaporised = $0.45 \times 24.985 = 11.243 \text{ kg/hr}$

- Water vaporised = $0.45 \times 1.315 = 0.592 \text{ kg/hr}$

$$Q = \dot{m} \times \lambda$$

- $Q_{\text{ethanol}} = (11.243/3600) \times 800 = 2.498 \text{ kW}$
- $Q_{\text{water}} = (0.592/3600) \times 2133 = 0.351 \text{ kW}$

$$Q_{\text{HX102 (cold)}} = 2.498 + 0.351 = 2.849 \text{ kW}$$

Hot Side (Reactor product: $360^\circ\text{C} \rightarrow 135^\circ\text{C}$):

- $\Delta T_{\text{hot}} = 360 - 135 = 225^\circ\text{C}$

$$Q = \dot{m} \times C_p \times \Delta T$$

$$Q = \dot{m} \times \lambda$$

- $Q_{\text{ethylene}} = (13.237/3600) \times 1.55 \times 225 = 1.284 \text{ kW [vapour } C_p]$
- $Q_{\text{water (sensible)}} = (9.815/3600) \times 2.01 \times 225 = 1.234 \text{ kW [vapour } C_p]$
- $Q_{\text{ethanol}} = (3.248/3600) \times 1.42 \times 225 = 0.289 \text{ kW [vapour } C_p]$
- $Q_{\text{water (latent, 0.7\% partial condensation near dew point)}} = (0.007 \times 9.815/3600) \times 2133 = 0.042 \text{ kW}$

$$Q_{\text{HX102 (hot)}} = 1.284 + 1.234 + 0.289 + 0.042 = 2.849 \text{ kW}$$

Heat balance check: Cold side = 2.849 kW, Hot side = 2.849 kW $\rightarrow$ Exact balance

5.5 Enthalpy Balance across Electric Heater:

The electric heater completes vaporisation of the remaining 55% liquid fraction and superheats the fully vapourised feed from 116°C to 450°C . Calculated in two parts:

Part A — Complete Vaporisation of Remaining Liquid at 116°C :

- Ethanol remaining liquid = $0.55 \times 24.985 = 13.742 \text{ kg/hr}$
- Water remaining liquid = $0.55 \times 1.315 = 0.723 \text{ kg/hr}$

$$Q = \dot{m} \times \lambda$$

- $Q_{\text{A ethanol}} = (13.742/3600) \times 800 = 3.053 \text{ kW}$
- $Q_{\text{A water}} = (0.723/3600) \times 2133 = 0.428 \text{ kW}$

$$Q_{\text{A}} = 3.053 + 0.428 = 3.481 \text{ kW}$$

Part B — Superheating Vapour from 116°C to 450°C:

- $\Delta T = 450 - 116 = 334^\circ\text{C}$

$$Q = \dot{m} \times C_p \times \Delta T$$

- $Q_{\text{ethanol}} = (24.985/3600) \times 1.42 \times 334 = 3.286 \text{ kW}$

- $Q_{\text{water}} = (1.315/3600) \times 2.01 \times 334 = 0.245 \text{ kW}$

$$Q_B = 3.286 + 0.245 = 3.531 \text{ kW}$$

$$Q_{\text{Heater}} = Q_A + Q_B = 3.481 + 3.531 = 7.012 \text{ kW (Heat IN, External)}$$

5.6 Enthalpy Balance across Adiabatic Fixed Bed Reactor:

The dehydration reaction is endothermic. In the adiabatic reactor, the heat of reaction is supplied entirely by the sensible heat of the feed, causing a temperature drop from 450°C to 360°C across the bed.

- Ethanol reacted = $24.985 - 3.248 = 21.737 \text{ kg/hr}$

- Moles reacted = $21.737 \times 1000 / 46 = 472.5 \text{ mol/hr}$

- $\Delta H_{\text{rxn}} = +45.5 \text{ kJ/mol}$

$$(\text{endothermic}) Q_{\text{rxn}} = \text{moles} \times$$

$$\Delta H_{\text{rxn}} / 3600$$

$$Q_{\text{rxn}} = 472.5 \times 45.5 / 3600 = 5.970 \text{ kW (absorbed by reaction)}$$

The reactor is equipped with a steam jacket to minimise heat loss to the surroundings and maintain near-adiabatic operating conditions. Steam at approximately 2 bar (saturation temperature ~120°C) is circulated through the jacket, providing an insulating thermal boundary around the reactor shell. Since the dehydration reaction is endothermic, the jacket does not serve as a cooling medium but rather prevents the reactor wall temperature from dropping below the reaction temperature, which could cause condensation of the feed vapour on the reactor walls and deactivation of the catalyst bed.

Steam Duty required: $Q = \dot{m}_{\text{steam}} \times \lambda$

At 2 bar pressure, saturation pressure of steam = 120.21°C, $\lambda = 2201.56 \text{ KJ/kg}$

$$Q = \dot{m}_{\text{steam}} \times \lambda$$

$$5.970 = \dot{m}_{\text{steam}} \times 2201.56$$

$$\dot{m}_{\text{steam}} = 5.970 \times 3600 / 2201.56 = 10 \text{ kg/hr}$$

5.7 Enthalpy Balance across Expander:

The expander reduces pressure from 4 bar to 2 bar, recovering mechanical work. An isentropic expansion model with efficiency is applied:

- $T_{\text{in}} = 40^\circ\text{C} = 313 \text{ K}$ (after HX101 hot outlet), $P_{\text{in}} = 4 \text{ bar}$, $P_{\text{out}} = 2 \text{ bar}$
- Mixture $C_p \approx 1.55 \text{ kJ/kg}\cdot\text{K}$ (ethylene dominant), $\gamma \approx 1.24$
- Isentropic efficiency $\eta = 0.75$ (assumed)

$$(P_2/P_1)^{((\gamma-1)/\gamma)} = (2/4)^{(0.24/1.24)} = (0.5)^{0.1935} = 0.8735$$

$$W = \dot{m} \times C_p \times T_{\text{in}} \times [1 - (P_2/P_1)^{((\gamma-1)/\gamma)}] \times \eta$$

$$= 7.306 \times 10^{-3} \times 1.55 \times 313 \times (1 - 0.8735) \times 0.75$$

$$W_{\text{expander}} = 0.440 \text{ kW (Work recovered, OUT)}$$

5.8 Enthalpy Balance across Refrigeration Cooler:

The refrigeration cooler cools the stream from 28.43°C to 8.43°C at 2 bar. At these temperatures, water and ethanol are fully in the liquid phase, so liquid C_p values are applied:

- $\Delta T = 28.43 - 8.43 = 20.00^\circ\text{C}$
- $Q_{\text{ethylene}} = (13.237/3600) \times 1.55 \times 20.00 = 0.114 \text{ kW [vapour } C_p]$
- $Q_{\text{water}} = (9.815/3600) \times 4.18 \times 20.00 = 0.228 \text{ kW [liquid } C_p]$
- $Q_{\text{ethanol}} = (3.248/3600) \times 2.44 \times 20.00 = 0.044 \text{ kW [liquid } C_p]$

$$Q_{\text{cooler}} = 0.114 + 0.228 + 0.044 = 0.386 \text{ kW (Heat removed, External)}$$

5.9 Enthalpy Balance across Flash Drum:

The flash drum operates adiabatically at 8.43°C and 2 bar. No external heat is added or removed.

- Feed enthalpy = Vapour outlet enthalpy + Liquid outlet enthalpy
- No external heat duty: $Q_{\text{flash}} = 0 \text{ kW}$

5.10 Overall Energy Balance Summary:

The overall energy balance across the complete flowsheet is summarised in Table 5.1 below. Internal heat exchange between process streams (HX 101 and HX 102) is not counted as external utility since no external energy source is involved.

Table 5.1: Overall Energy Balance Summary

Equipment	Duty (kW)	Type	Source
Feed Pump	0.004	Work IN	External
HX 101	1.872	Heat exchanged (balanced)	Internal
HX 102	2.849	Heat exchanged (balanced)	Internal
Electric Heater	7.012	Heat IN	External
Reactor	5.970	Endothermic (from feed)	Adiabatic
Expander	0.440	Work OUT	External
Refrigeration Cooler	0.386	Heat OUT	External
Flash Drum	0	Adiabatic	None

Net External Energy Demand:

- Energy IN: $Q_{\text{heater}} + W_{\text{pump}} = 7.012 + 0.004 = 7.016 \text{ kW}$
 - Energy OUT (recovered): $W_{\text{expander}} + Q_{\text{cooler}} = 0.440 + 0.386 = 0.826 \text{ kW}$
- Net External Demand = $7.016 - 0.826 = 6.190 \text{ kW}$

Table 5.2: Net External Energy Summary

Item	Value (kW)
Electric Heater Duty	7.012
Pump Work	0.004
Expander Work Recovered	0.440
Refrigeration Duty	0.386
Net External Energy Demand	6.190

The heat integration network (HX 101 and HX 102) recovers 1.872 kW and 2.849 kW respectively from the hot reactor effluent, heat that would otherwise need to be supplied externally. Both heat exchangers are thermodynamically balanced with exact cold-side to hot-side agreement, confirmed by applying phase-appropriate C_p values and accounting for partial condensation effects. The expander further offsets energy demand by recovering 0.440 kW of mechanical work from the pressure reduction step. The net external energy requirement of the plant is 6.190 kW.